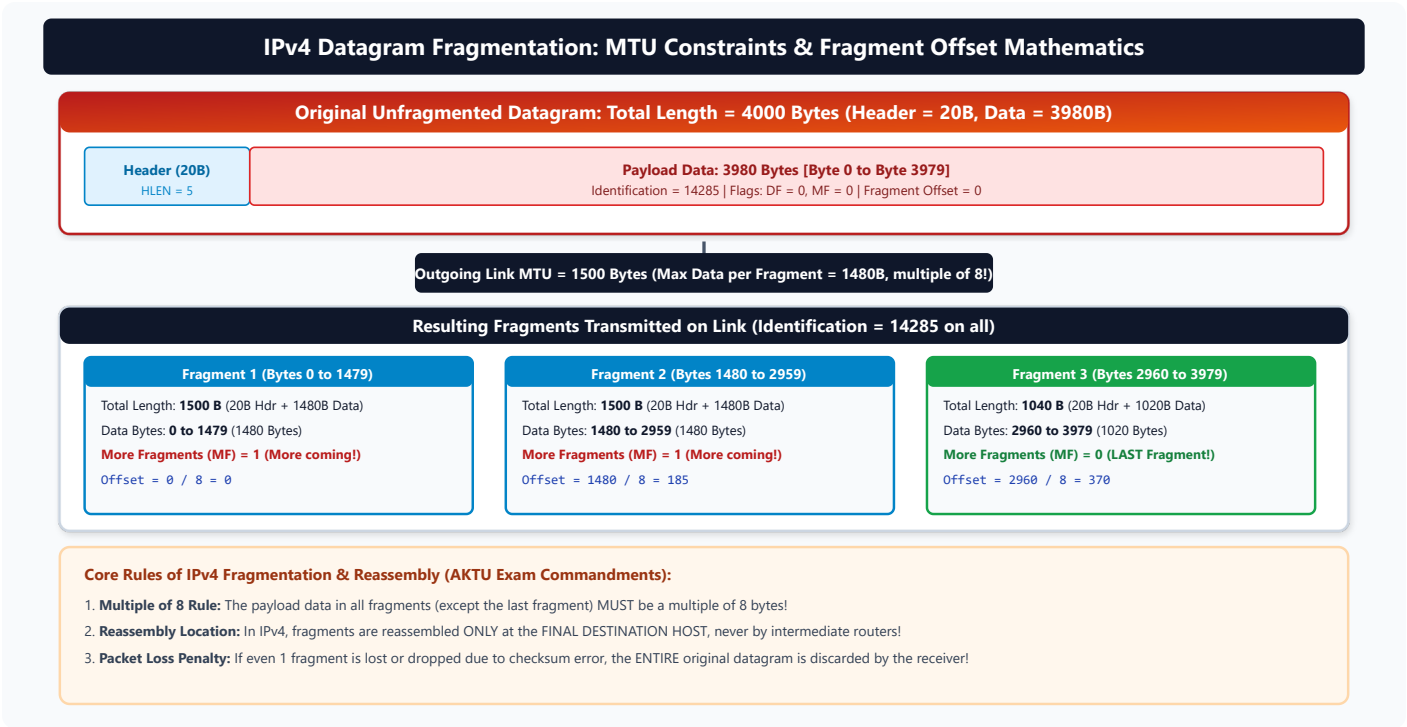


Module 03: IPv4 Fragmentation, Reassembly & Solved MTU Numericals

Unit 3: Network Layer

MTU & Offset Mathematics

Mathematical principles of IPv4 fragmentation across heterogeneous networks with differing MTUs, the 8-byte offset unit rule, destination-only reassembly, and end-to-end numerical derivations.



1. The MTU Boundary Constraint

When an IP datagram exceeds the Maximum Transmission Unit of an outgoing physical link, it is fragmented into smaller packets. Each fragment retains the original Identification number, an appropriate More Fragments (MF) flag, and a Fragment Offset in 8-byte increments.

2. Fragmentation Math Rules

Golden Fragmentation Laws

Max Data per Fragment = $\text{floor}((\text{MTU} - \text{Header}) / 8) \times 8$

Fragment Offset = Starting Byte Index / 8

MF = 1 for all fragments except the last fragment (MF = 0)